

Industry-Standard Dual Operational Amplifiers

1 Features

- Wide supply range of 3V to 36V (B, BA versions)
- Quiescent current: 300 μ A/ch (B, BA versions)
- Unity-gain bandwidth of 1.2MHz (B, BA versions)
- Common-mode input voltage range includes ground, enabling direct sensing near ground
- 2mV input offset voltage maximum at 25°C (BA version)
- 3mV input offset voltage maximum at 25°C (A, B versions)
- Internal RF and EMI filter (B, BA versions)
- On products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. On all other products, production processing does not necessarily include testing of all parameters.

2 Applications

- Merchant network and server power supply units
- Multi-function printers
- Power supplies and mobile chargers
- Motor control: AC induction, brushed DC, brushless DC, high-voltage, low-voltage, permanent magnet, and stepper motor
- Desktop PC and motherboard
- Indoor and outdoor air conditioners
- Washers, dryers, and refrigerators
- AC inverters, string inverters, central inverters, and voltage frequency drives
- Uninterruptible power supplies
- Electronic point-of-sale systems

3 Description

The LM358B and LM2904B devices are the next-generation versions of the industry-standard operational amplifiers (op amps) LM358 and LM2904, which include two high-voltage (36V) op amps. These

devices provide outstanding value for cost-sensitive applications, with features including low offset (300 μ V, typical), common-mode input range to ground, and high differential input voltage capability.

The LM358B and LM2904B op amps simplify circuit design with enhanced features such as unity-gain stability, lower offset voltage maximum of 3mV (2mV maximum for LM358BA and LM2904BA), and lower quiescent current of 300 μ A per amplifier (typical). High ESD (2kV, HBM) and integrated EMI and RF filters enable the LM358B and LM2904B devices to be used in the most rugged, environmentally challenging applications.

The LM358B and LM2904B amplifiers are available in micro-sized packaging, such as the SOT23-8, as well as industry standard packages including SOIC, TSSOP, and VSSOP.

Package Information

PART NUMBER ⁽¹⁾	PACKAGE	PACKAGE SIZE ⁽²⁾
LM358B, LM358BA, LM2904B, LM2904BA, LM358, LM358A, LM2904, LM2904V, LM258, LM258A	D (SOIC, 8)	4.9mm × 6mm
LM358B, LM358BA, LM2904B, LM2904BA, LM358, LM358A, LM2904, LM2490V	PW (TSSOP, 8)	3mm × 6.4mm
LM358B, LM358BA, LM2904B, LM2904BA, LM358, LM358A, LM2904, LM2904V, LM258, LM258A	DGK (VSSOP, 8)	3mm × 4.9mm
LM358B, LM358BA, LM2904B, LM2904BA	DDF (SOT-23, 8)	2.9mm × 2.8mm
LM358, LM2904	PS (SO, 8)	6.2mm × 7.8mm
LM358, LM2904, LM358A, LM258, LM258A	P (PDIP, 8)	9.81mm × 9.43mm
LM158, LM158A	JG (CDIP, 8)	9.6mm × 6.67mm
LM158, LM158A	FK (LCCC, 20)	8.89mm × 8.89mm

Family Comparison

Specification	LM358B LM358BA	LM2904B LM2904BA	LM358 LM358A	LM2904	LM2904V LM2904AV	LM258 LM258A	LM158 LM158A	Units
Supply voltage	3 to 36	3 to 36	3 to 30	3 to 26	3 to 30	3 to 30	3 to 30	V
Offset voltage (max, 25°C)	± 3 ± 2	± 3 ± 2	± 7 ± 3	± 7	± 7 ± 2	± 5 ± 3	± 5 ± 2	mV
Input bias current (typ / max)	10 / 35	10 / 35	20 / 250 15 / 100	20 / 250	20 / 250	20 / 150 15 / 80	20 / 150 15 / 50	nA
Gain bandwidth product	1.2	1.2	0.7	0.7	0.7	0.7	0.7	MHz
Supply current (typ, per channel)	0.3	0.3	0.35	0.35	0.35	0.35	0.35	mA
ESD (HBM)	2000	2000	500	500	500	500	500	V
Operating ambient temperature	–40 to 85	–40 to 125	0 to 70	–40 to 125	–40 to 125	–25 to 85	–55 to 125	°C

(1) For all available packages, see the orderable addendum at the end of the data sheet.

(2) The package size (length × width) is a nominal value and includes pins, where applicable.

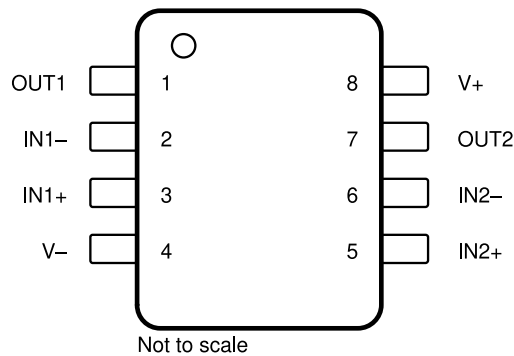


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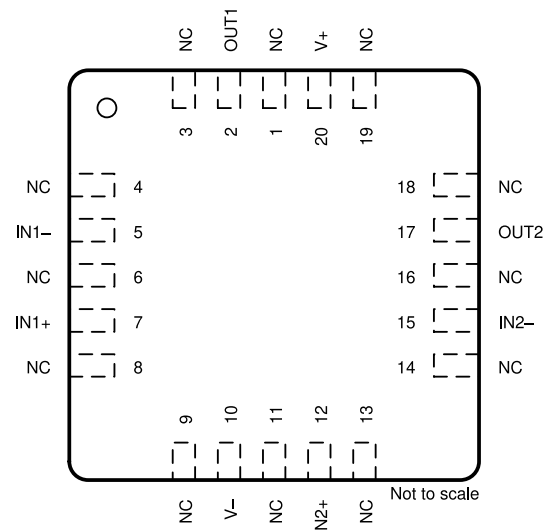
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4 Pin Configuration and Functions



**Figure 4-1. D, DDF, DGK, P, PS, PW, and JG
Package
8-Pin SOIC, SOT23-8, VSSOP, PDIP, SO, TSSOP,
and CDIP
Top View**



NC - No internal connection

**Figure 4-2. FK Package
20-Pin LCCC
Top View**

Table 4-1. Pin Functions

NAME	PIN		I/O	DESCRIPTION
	LCCC ⁽¹⁾	SOIC, SOT23-8, VSSOP, CDIP, PDIP, SO, TSSOP, CFP ⁽¹⁾		
IN1–	5	2	I	Negative input
IN1+	7	3	I	Positive input
IN2–	15	6	I	Negative input
IN2+	12	5	I	Positive input
OUT1	2	1	O	Output
OUT2	17	7	O	Output
V–	10	4	—	Negative (lowest) supply or ground (for single-supply operation)
NC	1, 3, 4, 6, 8, 9, 11, 13, 14, 16, 18, 19	—	—	No internal connection
V+	20	8	—	Positive (highest) supply

(1) For a listing of which devices are available in what packages, see [Section 3](#).

5 Specifications

5.1 Absolute Maximum Ratings

over operating ambient temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT	
Supply voltage, $V_S = ([V+] - [V-])$	LM358B, LM358BA, LM2904B, LM2904BA		±20 or 40	V	
	LM158, LM258, LM358, LM158A, LM258A, LM358A, LM2904V		±16 or 32		
	LM2904		±13 or 26		
Differential input voltage, V_{ID} ⁽²⁾	LM358B, LM358BA, LM2904B, LM2904BA,LM158, LM258, LM358, LM158A, LM258A, LM358A, LM2904V	–32	32	V	
	LM2904	–26	26		
Input voltage, V_I	Either input	LM358B, LM358BA, LM2904B, LM2904BA	–0.3	40	V
		LM158, LM258, LM358, LM158A, LM258A, LM358A, LM2904V	–0.3	32	
		LM2904	–0.3	26	
Duration of output short circuit (one amplifier) to ground at (or below) $T_A = 25^{\circ}\text{C}$, $V_S \leq 15\text{ V}$ ⁽³⁾			Unlimited	s	
Operating ambient temperature, T_A	LM158, LM158A	–55	125	$^{\circ}\text{C}$	
	LM258, LM258A	–25	85		
	LM358B, LM358BA	–40	85		
	LM358, LM358A	0	70		
	LM2904B, LM2904BA, LM2904, LM2904V	–40	125		
Operating virtual-junction temperature, T_J			150	$^{\circ}\text{C}$	
Storage temperature, T_{stg}		–65	150	$^{\circ}\text{C}$	

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) Differential voltages are at $IN+$, with respect to $IN-$.

(3) Short circuits from outputs to V_S can cause excessive heating and eventual destruction.

5.2 ESD Ratings

		VALUE	UNIT	
LM358B, LM358BA, LM2904B, AND LM2904BA				
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1500	
LM158, LM258, LM358, LM158, LM258A, LM358A, LM2904, AND LM2904V				
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±500	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1000	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

		MIN	MAX	UNIT
V _S	Supply voltage, V _S = ([V+] – [V–])	LM358B, LM358BA, LM2904B, LM2904BA	336	V
		LM158, LM258, LM358, LM158A, LM258A, LM358A, LM2904V	330	
		LM2904	326	
V _{CM}	Common-mode voltage	V–	V+ – 2	V
T _A	Operating ambient temperature	LM358B, LM358BA	–4085	°C
		LM2904B, LM2904BA, LM2904, LM2904V	–40125	
		LM358, LM358A	070	
		LM258, LM258A	–2085	
		LM158, LM158A	–55125	

5.4 Thermal Information

THERMAL METRIC ⁽¹⁾		LM258, LM258A, LM358, LM358A, LM358B, LM358BA, LM2904, LM2904B, LM2904BA, LM2904V ⁽²⁾						LM158, LM158A		UNIT
		D (SOIC)	DGK (VSSOP)	P (PDIP)	PS (SO)	PW (TSSOP)	DDF (SOT-23)	FK (LCCC)	JG (CDIP)	
		8 PINS	8 PINS	8 PINS	8 PINS	8 PINS	8PINS	20 PINS	8 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	124.7	181.4	80.9	116.9	171.7	164.3	84.0	112.4	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	66.9	69.4	70.4	62.5	68.8	98.1	56.9	63.6	°C/W
R _{θJB}	Junction-to-board thermal resistance	67.9	102.9	57.4	68.6	99.2	82.1	57.5	100.3	°C/W
ψ _{JT}	Junction-to-top characterization parameter	19.2	11.8	40	21.9	11.5	11.4	51.7	35.7	°C/W
ψ _{JB}	Junction-to-board characterization parameter	67.2	101.2	56.9	67.6	97.9	81.7	57.1	93.3	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	—	—	—	—	—	—	10.6	22.3	°C/W

- (1) For more information about traditional and new thermal metrics, see [Semiconductor and IC Package Thermal Metrics](#).
(2) For a listing of which devices are available in what packages, see [Section 3](#).

5.5 Electrical Characteristics: LM358B and LM358BA

$V_S = (V+) - (V-) = 5\text{ V} - 36\text{ V} (\pm 2.5\text{ V} - \pm 18\text{ V})$, $T_A = 25^\circ\text{C}$, $V_{CM} = V_{OUT} = V_S / 2$, $R_L = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted)

PARAMETER			TEST CONDITIONS		MIN	TYP	MAX	UNIT
OFFSET VOLTAGE								
V _{OS}	Input offset voltage	LM358B				±0.3	±3.0	mV
				T _A = −40°C to +85°C			±4	mV
		LM358BA					±2.0	mV
				T _A = −40°C to +85°C			±2.5	mV
dV _{OS} /dT	Input offset voltage drift				T _A = −40°C to +85°C ⁽¹⁾	±3.5	11	µV/°C
PSRR	Power supply rejection ratio					±2	15	µV/V
	Channel separation, dc	f = 1 kHz to 20 kHz				±1		µV/V
INPUT VOLTAGE RANGE								
V _{CM}	Common-mode voltage range	V _S = 3 V to 36 V			(V−)	(V+) − 1.5		V
		V _S = 5 V to 36 V		T _A = −40°C to +85°C	(V−)	(V+) − 2		V
CMRR	Common-mode rejection ratio	(V−) ≤ V _{CM} ≤ (V+) − 1.5 V	V _S = 3 V to 36 V			20	100	µV/V
		(V−) ≤ V _{CM} ≤ (V+) − 2.0 V	V _S = 5 V to 36 V	T _A = −40°C to +85°C		25	316	
INPUT BIAS CURRENT								
I _B	Input bias current					−10	−35	nA
				T _A = −40°C to +85°C ⁽¹⁾			−50	nA
I _{OS}	Input offset current					0.5	4	nA
				T _A = −40°C to +85°C ⁽¹⁾			5	nA
dI _{OS} /dT	Input offset current drift			T _A = −40°C to +85°C		10		pA/°C
NOISE								
E _n	Input voltage noise	f = 0.1 to 10 Hz				3		µV _{PP}
e _n	Input voltage noise density	f = 1 kHz				40		nV/√Hz
INPUT IMPEDANCE								
Z _{ID}	Differential					10 0.1		MΩ pF
Z _{IC}	Common-mode					4 1.5		GΩ pF
OPEN-LOOP GAIN								
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 10 kΩ, connected to (V−)			70	140		V/mV
				T _A = −40°C to +85°C	35			V/mV
FREQUENCY RESPONSE								
GBW	Gain bandwidth product					1.2		MHz
SR	Slew rate	G = + 1				0.5		V/µs
Θ _m	Phase margin	G = + 1, R _L = 10kΩ, C _L = 20 pF				56		°
t _{OR}	Overload recovery time	V _{IN} × gain > V _S				10		µs
t _s	Settling time	To 0.1%, V _S = 5 V, 2-V step , G = +1, C _L = 100 pF				4		µs
THD+N	Total harmonic distortion + noise	G = + 1, f = 1 kHz, V _O = 3.53 V _{RMS} , V _S = 36 V, R _L = 100k, I _{OUT} ≤ ±50 µA, BW = 80 kHz				0.001		%
OUTPUT								
V _O	Voltage output swing from rail	Positive rail (V+)		I _{OUT} = 50 µA	1.35	1.42		V
				I _{OUT} = 1 mA	1.4	1.48		V
				I _{OUT} = 5 mA ⁽¹⁾	1.5	1.61		V
		Negative rail (V−)		I _{OUT} = 50 µA	100	150		mV
				I _{OUT} = 1 mA	0.75	1		V
				V _S = 5 V, R _L ≤ 10 kΩ connected to (V−)	T _A = −40°C to +85°C	5	20	
I _O	Output current	V _S = 15 V; V _O = V−; V _{ID} = 1 V	Source ⁽¹⁾		−20	−30	mA	
				T _A = −40°C to +85°C	−10			
		V _S = 15 V; V _O = V+; V _{ID} = −1 V	Sink ⁽¹⁾		10	20		
				T _A = −40°C to +85°C	5			
		V _{ID} = −1 V; V _O = (V−) + 200 mV		60	100		µA	
I _{SC}	Short-circuit current	V _S = 20 V, (V+) = 10 V, (V−) = −10 V, V _O = 0 V				±40	±60	mA
C _{LOAD}	Capacitive load drive					100		pF
R _O	Open-loop output resistance	f = 1 MHz, I _O = 0 A				300		Ω

5.5 Electrical Characteristics: LM358B and LM358BA (continued)

$V_S = (V+) - (V-) = 5\text{ V} - 36\text{ V} (\pm 2.5\text{ V} - \pm 18\text{ V})$, $T_A = 25^\circ\text{C}$, $V_{CM} = V_{OUT} = V_S / 2$, $R_L = 10\text{k}$ connected to $V_S / 2$
(unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER SUPPLY						
I_Q	Quiescent current per amplifier	$V_S = 5\text{ V}$; $I_O = 0\text{ A}$		300	460	μA
I_Q	Quiescent current per amplifier	$V_S = 36\text{ V}$; $I_O = 0\text{ A}$				

(1) Specified by characterization only.

5.6 Electrical Characteristics: LM2904B and LM2904BA

$V_S = (V_+) - (V_-) = 5\text{ V} - 36\text{ V} (\pm 2.5\text{ V} - \pm 18\text{ V})$, $T_A = 25^\circ\text{C}$, $V_{CM} = V_{OUT} = V_S/2$, $R_L = 10\text{ k}\Omega$ connected to $V_S/2$
(unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
OFFSET VOLTAGE							
V _{OS}	Input offset voltage	LM2904B			±0.3	±3.0	mV
			T _A = −40°C to +125°C		±4	mV	
		LM2904BA			±2.0		mV
			T _A = −40°C to +125°C		±3.0	mV	
dV _{OS} /dT	Input offset voltage drift		T _A = −40°C to +125°C ⁽¹⁾	±3.5	12	μV/°C	
PSRR	Power Supply Rejection Ratio				±2	15	μV/V
	Channel separation, dc	f = 1 kHz to 20 kHz			±1		μV/V
INPUT VOLTAGE RANGE							
V _{CM}	Common-mode voltage range	V _S = 3 V to 36 V			(V−)	(V+) − 1.5	V
		V _S = 5 V to 36 V		T _A = −40°C to +125°C	(V−)	(V+) − 2	V
CMRR	Common-mode rejection ratio	(V−) ≤ V _{CM} ≤ (V+) − 1.5 V	V _S = 3 V to 36 V		20	100	μV/V
		(V−) ≤ V _{CM} ≤ (V+) − 2.0 V	V _S = 5 V to 36 V	T _A = −40°C to +125°C	25	316	
INPUT BIAS CURRENT							
I _B	Input bias current				−10	−35	nA
				T _A = −40°C to +125°C ⁽¹⁾		−50	nA
I _{OS}	Input offset current				0.5	4	nA
				T _A = −40°C to +125°C ⁽¹⁾		5	nA
dI _{OS} /dT	Input offset current drift			T _A = −40°C to +125°C	10		pA/°C
NOISE							
E _n	Input voltage noise	f = 0.1 to 10 Hz			3		μV _{PP}
e _n	Input voltage noise density	f = 1 kHz			40		nV/√Hz
INPUT IMPEDANCE							
Z _{ID}	Differential				10 0.1		MΩ pF
Z _{IC}	Common-mode				4 1.5		GΩ pF
OPEN-LOOP GAIN							
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 10 kΩ, connected to (V−)			70	140	V/mV
				T _A = −40°C to +125°C	35		V/mV
FREQUENCY RESPONSE							
GBW	Gain bandwidth product				1.2		MHz
SR	Slew rate	G = + 1			0.5		V/μs
Φ _m	Phase margin	G = + 1, R _L = 10kΩ, C _L = 20 pF			56		°
t _{OR}	Overload recovery time	V _{IN} × gain > V _S			10		μs
t _s	Settling time	To 0.1%, V _S = 5 V, 2-V Step , G = +1, C _L = 100 pF			4		μs
THD+N	Total harmonic distortion + noise	G = + 1, f = 1 kHz, V _O = 3.53 V _{RMS} , V _S = 36V, R _L = 100k, I _{OUT} ≤ ±50μA, BW = 80 kHz			0.001		%
OUTPUT							
V _O	Voltage output swing from rail	Positive Rail (V+)		I _{OUT} = 50 μA	1.35	1.42	V
				I _{OUT} = 1 mA	1.4	1.48	V
				I _{OUT} = 5 mA ⁽¹⁾	1.5	1.61	V
		Negative Rail (V-)		I _{OUT} = 50 μA	100	150	mV
				I _{OUT} = 1 mA	0.75	1	V
				V _S = 5 V, R _L ≤ 10 kΩ connected to (V−)	T _A = −40°C to +125°C	5	20
I _O	Output current	V _S = 15 V; V _O = V−; V _{ID} = 1 V	Source ⁽¹⁾		−20	−30	mA
			T _A = −40°C to +125°C	−10			
		V _S = 15 V; V _O = V+; V _{ID} = −1 V	Sink ⁽¹⁾		10	20	
			T _A = −40°C to +125°C	5			
		V _{ID} = −1 V; V _O = (V−) + 200 mV			60	100	μA
I _{SC}	Short-circuit current	V _S = 20 V, (V+) = 10 V, (V−) = −10 V, V _O = 0 V			±40	±60	mA
C _{LOAD}	Capacitive load drive				100		pF
R _O	Open-loop output resistance	f = 1 MHz, I _O = 0 A			300		Ω
POWER SUPPLY							

$V_S = (V+) - (V-) = 5\text{ V} - 36\text{ V} (\pm 2.5\text{ V} - \pm 18\text{ V})$, $T_A = 25^\circ\text{C}$, $V_{CM} = V_{OUT} = V_S/2$, $R_L = 10\text{k}$ connected to $V_S/2$
(unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_Q	Quiescent current per amplifier	$V_S = 5\text{ V}$; $I_O = 0\text{ A}$	$T_A = -40^\circ\text{C}$ to $+125^\circ\text{C}$		300	460	μA
I_Q	Quiescent current per amplifier	$V_S = 36\text{ V}$; $I_O = 0\text{ A}$				800	μA

(1) Specified by characterization only

5.7 Electrical Characteristics: LM358, LM358A

For $V_S = (V+) - (V-) = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS ⁽¹⁾		MIN	TYP ⁽²⁾	MAX	UNIT
OFFSET VOLTAGE							
V _{OS}	Input offset voltage	V _S = 5 V to 30 V; V _{C M} = 0 V; V _O = 1.4 V	LM358		3	7	mV
				T _A = 0°C to 70°C		9	
			LM358A		2	3	
				T _A = 0°C to 70°C		5	
dV _{OS} /dT	Input offset voltage drift		LM358	T _A = 0°C to 70°C	7		μV/°C
			LM358A	T _A = 0°C to 70°C	7	20	
PSRR	Input offset voltage vs power supply (ΔV _{IO} /ΔV _S)	V _S = 5 V to 30 V		65	100		dB
V _{O1} / V _{O2}	Channel separation	f = 1 kHz to 20 kHz			120		dB
INPUT VOLTAGE RANGE							
V _{CM}	Common-mode voltage range	V _S = 5 V to 30 V	LM358		(V–)	(V+) – 1.5	V
		V _S = 30 V	LM358A				
		V _S = 5 V to 30 V	LM358	T _A = 0°C to 70°C	(V–)	(V+) – 2	
		V _S = 30 V					
CMRR	Common-mode rejection ratio	V _S = 5 V to 30 V; V _{CM} = 0 V		65	80		dB
INPUT BIAS CURRENT							
I _B	Input bias current	V _O = 1.4 V	LM358		–20	–250	nA
				T _A = 0°C to 70°C		–500	
			LM358A		–15	–100	
				T _A = 0°C to 70°C		–200	
I _{OS}	Input offset current	V _O = 1.4 V	LM358		2	50	nA
				T _A = 0°C to 70°C		150	
			LM358A		2	30	
				T _A = 0°C to 70°C		75	
dI _{OS} /dT	Input offset current drift		LM358A	T _A = 0°C to 70°C	10	300	pA/°C
NOISE							
e _n	Input voltage noise density	f = 1 kHz			40		nV/√ Hz
OPEN-LOOP GAIN							
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 2 kΩ		25	100	V/mV	
			T _A = 0°C to 70°C	15			
FREQUENCY RESPONSE							
GBW	Gain bandwidth product				0.7		MHz
SR	Slew rate	G = +1			0.3		V/μs
OUTPUT							
V _O	Voltage output swing from rail	Positive rail	V _S = 30 V; R _L = 2 kΩ	T _A = 0°C to 70°C	4		V
			V _S = 30 V; R _L ≥ 10 kΩ		2	3	
			V _S = 5 V; R _L ≥ 2 kΩ			1.5	
		Negative rail	V _S = 5 V; R _L ≤ 10 kΩ	T _A = 0°C to 70°C	5	20	mV
I _O	Output current	V _S = 15 V; V _O = 0 V; V _{ID} = 1 V	Source		–20	–30	mA
				LM358A	–60		
				T _A = 0°C to 70°C	–10		
		Sink		10	20		
			T _A = 0°C to 70°C	5			
V _{ID} = –1 V; V _O = 200 mV		12	30		μA		
I _{SC}	Short-circuit current	V _S = 10 V; V _O = V _S / 2			±40	±60	mA
POWER SUPPLY							
I _Q	Quiescent current per amplifier	V _O = 2.5 V; I _O = 0 A		T _A = 0°C to 70°C	350	600	μA
		V _S = 30 V; V _O = 15 V; I _O = 0 A			500	1000	

- (1) All characteristics are measured under open-loop conditions, with zero common-mode input voltage, unless otherwise specified. Maximum V_S for testing purposes is 30 V for LM358 and LM358A.
- (2) All typical values are $T_A = 25^\circ\text{C}$.

5.8 Electrical Characteristics: LM2904, LM2904V

For $V_S = (V^+) - (V^-) = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS ⁽¹⁾		MIN		TYP ⁽²⁾		MAX	UNIT
OFFSET VOLTAGE									
V _{OS}	Input offset voltage	V _S = 5 V to maximum; V _{C M} = 0 V; V _O = 1.4 V	Non-A suffix devices		3		7	mV	
				T _A = −40°C to 125°C	10				
			A-suffix devices		1		2		
				T _A = −40°C to 125°C	4				
dV _{OS} /dT	Input offset voltage drift			T _A = −40°C to 125°C	7			μV/°C	
PSRR	Input offset voltage vs power supply (ΔV _{IO} /ΔV _S)	V _S = 5 V to 30 V			65	100		dB	
V _{O1} / V _{O2}	Channel separation	f = 1 kHz to 20 kHz			120			dB	
INPUT VOLTAGE RANGE									
V _{CM}	Common-mode voltage range	V _S = 5 V to maximum		(V−)	(V+) − 1.5		V		
			T _A = −40°C to 125°C	(V−)	(V+) − 2				
CMRR	Common-mode rejection ratio	V _S = 5 V to maximum; V _{CM} = 0 V			65	80		dB	
INPUT BIAS CURRENT									
I _B	Input bias current	V _O = 1.4 V			−20	−250	nA		
			T _A = −40°C to 125°C		−500				
I _{OS}	Input offset current	V _O = 1.4 V	Non-V suffix device		2	50	nA		
				T _A = −40°C to 125°C		300			
			V-suffix device		2	50			
				T _A = −40°C to 125°C		150			
dI _{OS} /dT	Input offset current drift			T _A = −40°C to 125°C	10		pA/°C		
NOISE									
e _n	Input voltage noise density	f = 1 kHz			40			nV/√ Hz	
OPEN-LOOP GAIN									
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 2 kΩ		25	100	V/mV			
			T _A = −40°C to 125°C	15					
FREQUENCY RESPONSE									
GBW	Gain bandwidth product				0.7			MHz	
SR	Slew rate	G = +1			0.3			V/μs	
OUTPUT									
V _O	Voltage output swing from rail	Positive rail	R _L ≥ 10 kΩ		V _S − 1.5		V		
			Non-V suffix device	V _S = maximum; R _L = 2 kΩ		4			
				V _S = maximum; R _L ≥ 10 kΩ		2		3	
			V-suffix device	V _S = maximum; R _L = 2 kΩ		6			
		V _S = maximum; R _L ≥ 10 kΩ			4	5			
			Negative rail	V _S = 5 V; R _L ≤ 10 kΩ	T _A = −40°C to 125°C	5		20	mV
I _O	Output current	V _S = 15 V; V _O = 0 V; V _{ID} = 1 V	Source		−20	−30	mA		
			T _A = −40°C to 125°C		−10				
		V _S = 15 V; V _O = 15 V; V _{ID} = −1 V	Sink		10	20			
			T _A = −40°C to 125°C		5				
		V _{ID} = -1 V; V _O = 200 mV	Non-V suffix device		30		μA		
	V-suffix device		12	40					
I _{SC}	Short-circuit current	V _S = 10 V; V _O = V _S / 2			±40	±60	mA		
POWER SUPPLY									
I _Q	Quiescent current per amplifier	V _O = 2.5 V; I _O = 0 A		T _A = −40°C to 125°C	350	600	μA		
		V _S = maximum; V _O = maximum / 2; I _O = 0 A			500	1000			

- (1) All characteristics are measured under open-loop conditions, with zero common-mode input voltage, unless otherwise specified. Maximum V_S for testing purposes is 26 V for LM2904 and 32 V for LM2904V.
- (2) All typical values are $T_A = 25^\circ\text{C}$.

5.9 Electrical Characteristics: LM158, LM158A

For $V_S = (V_+) - (V_-) = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS ⁽¹⁾			MIN		TYP ⁽²⁾		MAX		UNIT	
OFFSET VOLTAGE												
V _{OS}	Input offset voltage	V _S = 5 V to 30 V; V _{C M} = 0 V; V _O = 1.4 V	LM158		3		5		mV			
				T _A = –55°C to 125°C	7							
			LM158A		2		4					
				T _A = –55°C to 125°C	7		15 ⁽³⁾					
dV _{OS} /dT	Input offset voltage drift		LM158	T _A = –55°C to 125°C	7				μV/°C			
			LM158A	T _A = –55°C to 125°C	7							
PSRR	Input offset voltage vs power supply (ΔV _{IO} /ΔV _S)	V _S = 5 V to 30 V			65		100		dB			
V _{O1} / V _{O2}	Channel separation	f = 1 kHz to 20 kHz			120				dB			
INPUT VOLTAGE RANGE												
V _{CM}	Common-mode voltage range	V _S = 5 V to 30 V	LM158		(V–)	(V+) – 1.5		V				
		V _S = 30 V	LM158A									
		V _S = 5 V to 30 V	LM158	T _A = –55°C to 125°C	(V–)	(V+) – 2						
		V _S = 30 V	LM158A									
CMRR	Common-mode rejection ratio	V _S = 5 V to 30 V; V _{CM} = 0 V			70		80		dB			
INPUT BIAS CURRENT												
I _B	Input bias current	V _O = 1.4 V	LM158		–20		–150		nA			
				T _A = –55°C to 125°C	–300							
			LM158A		–15		–50					
				T _A = –55°C to 125°C	–100							
I _{OS}	Input offset current	V _O = 1.4 V	LM158		2		30		nA			
				T _A = –55°C to 125°C	100							
			LM158A		2		10					
				T _A = –55°C to 125°C	30							
dI _{OS} /dT	Input offset current drift				10				pA/°C			
		LM158A	T _A = –55°C to 125°C	200								
NOISE												
e _n	Input voltage noise density	f = 1 kHz			40				nV/√ Hz			
OPEN-LOOP GAIN												
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 2 kΩ			50		100		V/mV			
			T _A = –55°C to 125°C	25								
FREQUENCY RESPONSE												
GBW	Gain bandwidth product				0.7				MHz			
SR	Slew rate	G = +1			0.3				V/μs			
OUTPUT												
V _O	Voltage output swing from rail	Positive rail	V _S = 30 V; R _L = 2 kΩ	T _A = –55°C to 125°C	4				V			
			V _S = 30 V; R _L ≥ 10 kΩ	2		3						
			V _S = 5 V; R _L ≥ 2 kΩ	1.5								
			Negative rail	V _S = 5 V; R _L ≤ 10 kΩ	T _A = –55°C to 125°C	5		20		mV		
I _O	Output current	V _S = 15 V; V _O = 0 V; V _{ID} = 1 V	Source		–20		–30		mA			
				LM158A	–60							
				T _A = –55°C to 125°C	–10							
		V _S = 15 V; V _O = 15 V; V _{ID} = –1 V	Sink		10		20					
				T _A = –55°C to 125°C	5							
	V _{ID} = –1 V; V _O = 200 mV				12		30		μA			
I _{SC}	Short-circuit current	V _S = 10 V; V _O = V _S / 2			±40		±60		mA			

5.9 Electrical Characteristics: LM158, LM158A (continued)

For $V_S = (V+) - (V-) = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS ⁽¹⁾		MIN	TYP ⁽²⁾	MAX	UNIT
POWER SUPPLY							
I _Q	Quiescent current per amplifier	V _O = 2.5 V; I _O = 0 A	T _A = −55°C to 125°C		350	600	μA
		V _S = 30 V; V _O = 15 V; I _O = 0 A			500	1000	

- (1) All characteristics are measured under open-loop conditions, with zero common-mode input voltage, unless otherwise specified. Maximum V_S for testing purposes is 30 V for LM158 and LM158A.
- (2) All typical values are $T_A = 25^\circ\text{C}$.
- (3) On products compliant to MIL-PRF-38535, this parameter is not production tested.

5.10 Electrical Characteristics: LM258, LM258A

For $V_S = (V+) - (V-) = 5\text{ V}$, $T_A = 25^\circ\text{C}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS ⁽¹⁾		MIN	TYP ⁽²⁾	MAX	UNIT	
OFFSET VOLTAGE								
V _{OS}	Input offset voltage	V _S = 5 V to 30 V; V _{C M} = 0 V; V _O = 1.4 V	LM258		3	5	mV	
				T _A = –25°C to 85°C		7		
			LM258A		2	3		
				T _A = –25°C to 85°C		4		
dV _{OS} /dT	Input offset voltage drift		LM258	T _A = –25°C to 85°C	7		μV/°C	
			LM258A		7	15		
PSRR	Input offset voltage vs power supply (ΔV _{IO} /ΔV _S)	V _S = 5 V to 30 V			65	100	dB	
V _{O1} / V _{O2}	Channel separation	f = 1 kHz to 20 kHz			120		dB	
INPUT VOLTAGE RANGE								
V _{CM}	Common-mode voltage range	V _S = 5 V to 30 V	LM258		(V–)	(V+) – 1.5	V	
		V _S = 30 V	LM258A					
		V _S = 5 V to 30 V	LM258	T _A = –25°C to 85°C	(V–)	(V+) – 2		
		V _S = 30 V	LM258A					
CMRR	Common-mode rejection ratio	V _S = 5 V to 30 V; V _{CM} = 0 V			70	80	dB	
INPUT BIAS CURRENT								
I _B	Input bias current	V _O = 1.4 V	LM258		–20	–150	nA	
				T _A = –25°C to 85°C		–300		
			LM258A		–15	–80		
				T _A = –25°C to 85°C		–100		
I _{OS}	Input offset current	V _O = 1.4 V	LM258		2	30	nA	
				T _A = –25°C to 85°C		100		
			LM258A		2	15		
				T _A = –25°C to 85°C		30		
dI _{OS} /dT	Input offset current drift				10		pA/°C	
			LM258A	T _A = –25°C to 85°C		200		
NOISE								
e _n	Input voltage noise density	f = 1 kHz			40		nV/√ Hz	
OPEN-LOOP GAIN								
A _{OL}	Open-loop voltage gain	V _S = 15 V; V _O = 1 V to 11 V; R _L ≥ 2 kΩ			50	100	V/mV	
				T _A = –25°C to 85°C	25			
FREQUENCY RESPONSE								
GBW	Gain bandwidth product				0.7		MHz	
SR	Slew rate	G = +1			0.3		V/μs	
OUTPUT								
V _O	Voltage output swing from rail	Positive rail	V _S = 30 V; R _L = 2 kΩ	T _A = –25°C to 85°C	4		V	
			V _S = 30 V; R _L ≥ 10 kΩ		2	3		
			V _S = 5 V; R _L ≥ 2 kΩ			1.5		
			Negative rail	V _S = 5 V; R _L ≤ 10 kΩ	T _A = –25°C to 85°C	5	20	mV
I _O	Output current	V _S = 15 V; V _O = 0 V; V _{ID} = 1 V	Source		–20	–30	mA	
			LM258A		–60			
		V _S = 15 V; V _O = 15 V; V _{ID} = –1 V	Sink		T _A = –25°C to 85°C	–10		20
					T _A = –25°C to 85°C	5		
	V _{ID} = –1 V; V _O = 200 mV			12	30	μA		
I _{SC}	Short-circuit current	V _S = 10 V; V _O = V _S / 2			±40	±60	mA	
POWER SUPPLY								
I _Q	Quiescent current per amplifier	V _O = 2.5 V; I _O = 0 A	T _A = –25°C to 85°C		350	600	μA	
		V _S = 30 V; V _O = 15 V; I _O = 0 A			500	1000		

- (1) All characteristics are measured under open-loop conditions, with zero common-mode input voltage, unless otherwise specified. Maximum V_S for testing purposes is 30 V for LM258 and LM258A.
(2) All typical values are $T_A = 25^\circ\text{C}$.

5.11 Typical Characteristics: LM358B and LM2904B

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

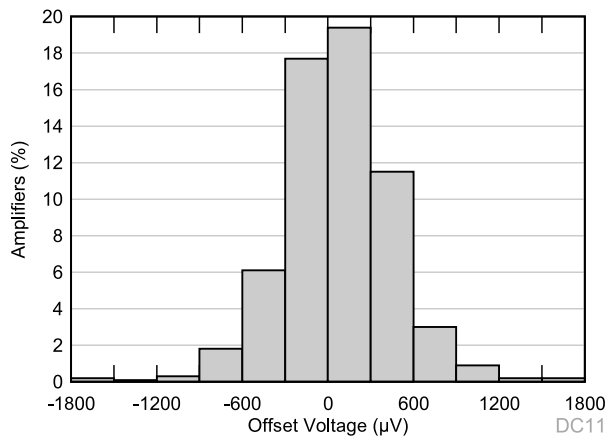


Figure 5-1. Offset Voltage Production Distribution

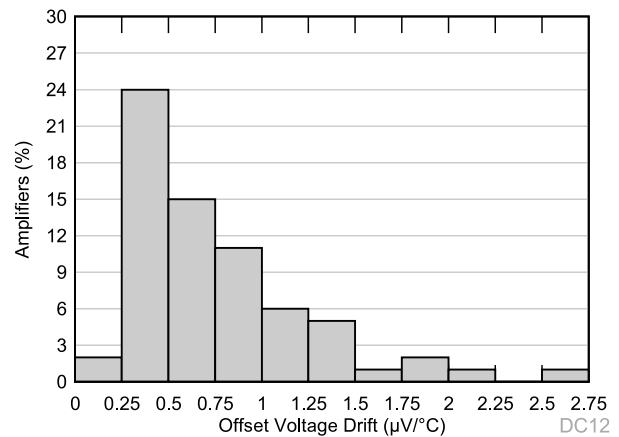


Figure 5-2. Offset Voltage Drift Distribution

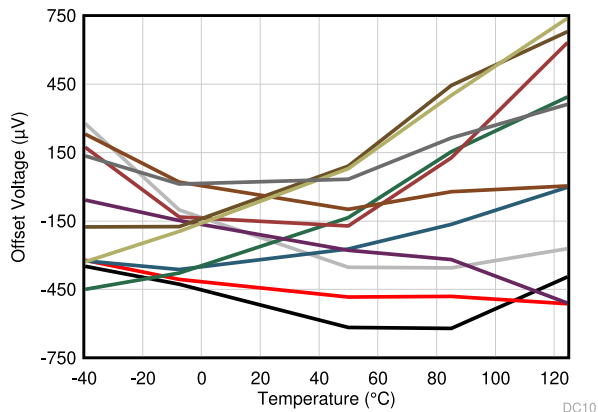


Figure 5-3. Offset Voltage vs Temperature

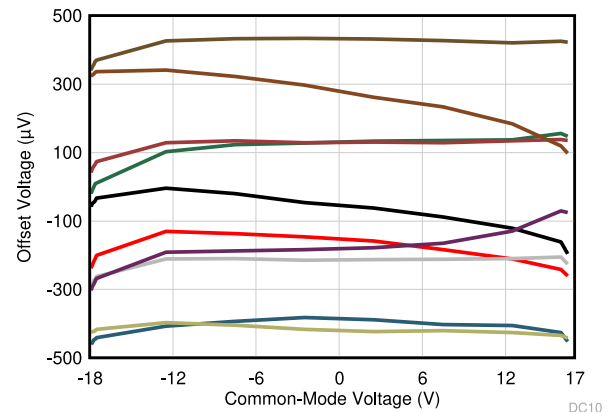


Figure 5-4. Offset Voltage vs Common-Mode Voltage

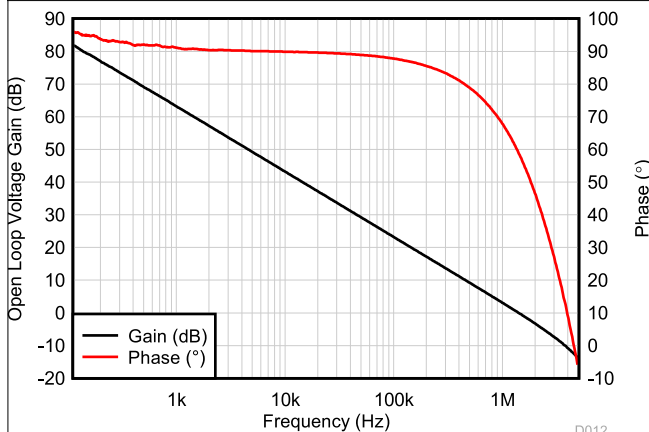


Figure 5-5. Open-Loop Gain and Phase vs Frequency

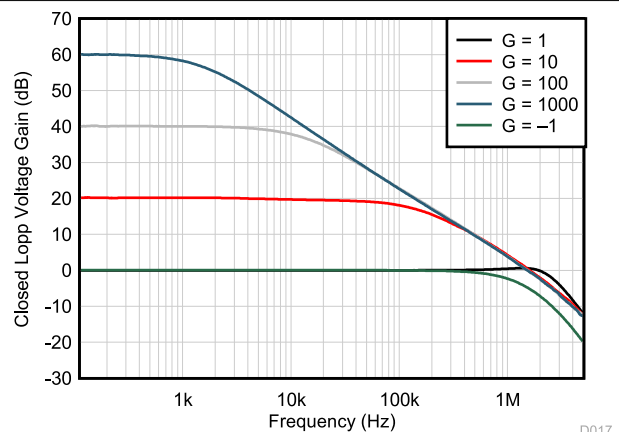


Figure 5-6. Closed-Loop Gain vs Frequency

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

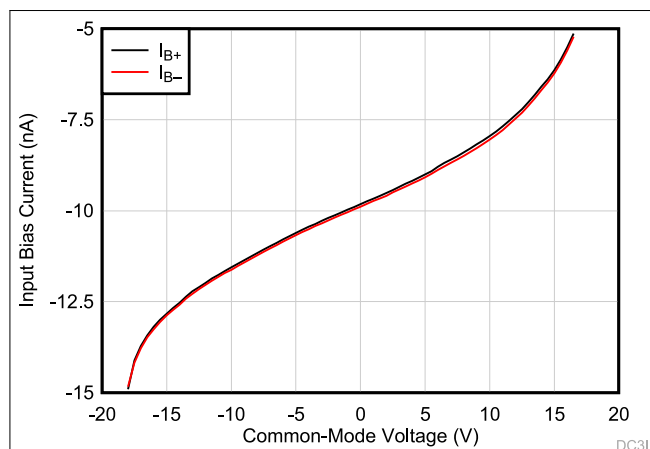


Figure 5-7. Input Bias Current vs Common-Mode Voltage

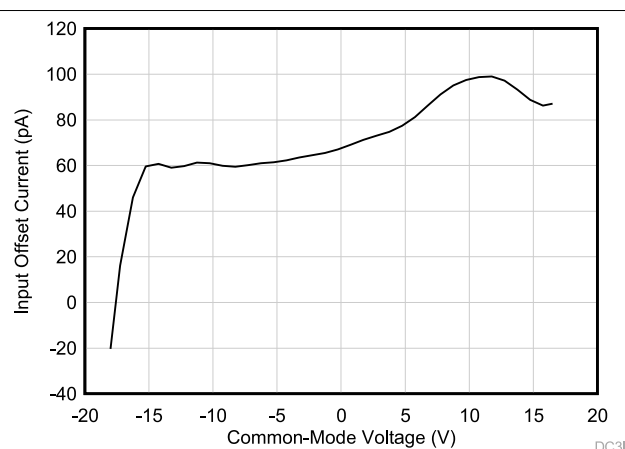


Figure 5-8. Input Offset Current vs Common-Mode Voltage

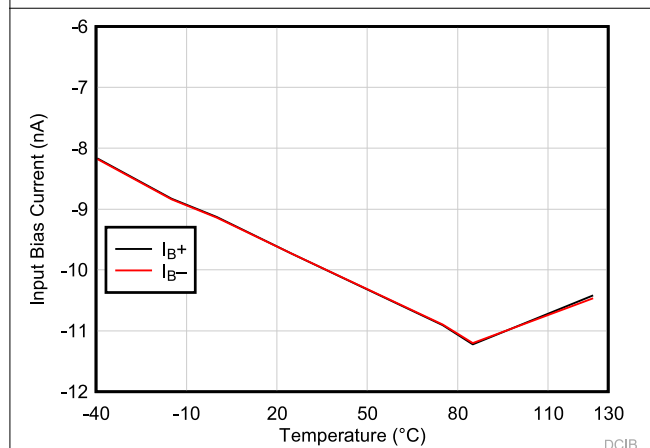


Figure 5-9. Input Bias Current vs Temperature

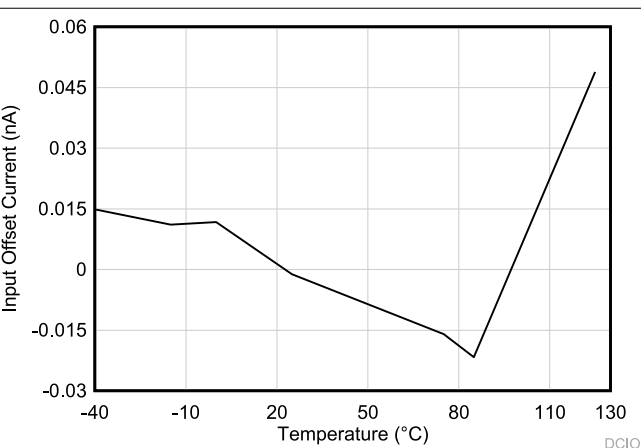


Figure 5-10. Input Offset Current vs Temperature

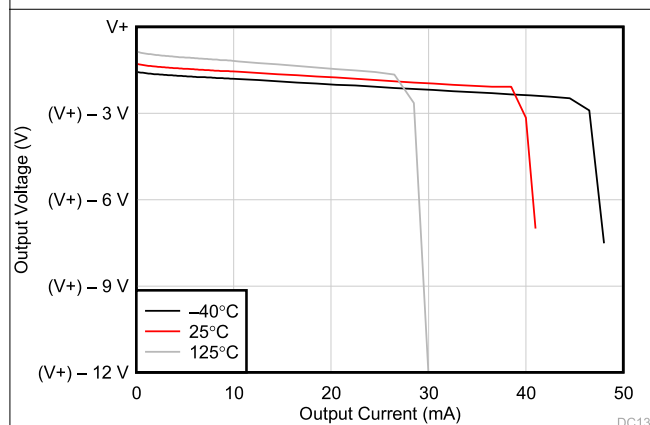


Figure 5-11. Output Voltage Swing vs Output Current (Sourcing)

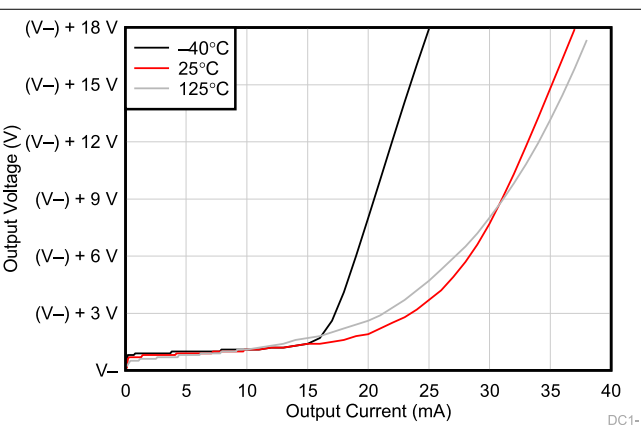


Figure 5-12. Output Voltage Swing vs Output Current (Sinking)

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

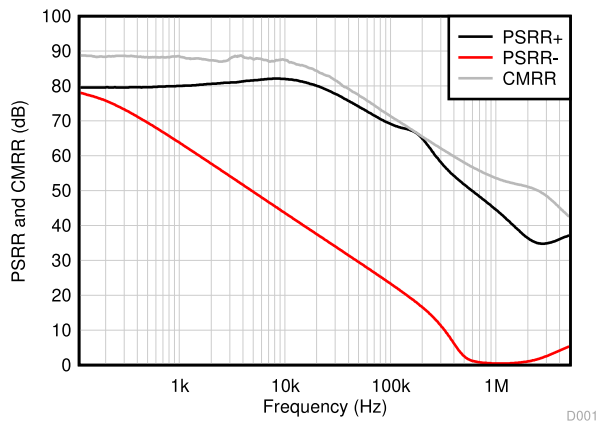


Figure 5-13. CMRR and PSRR vs Frequency

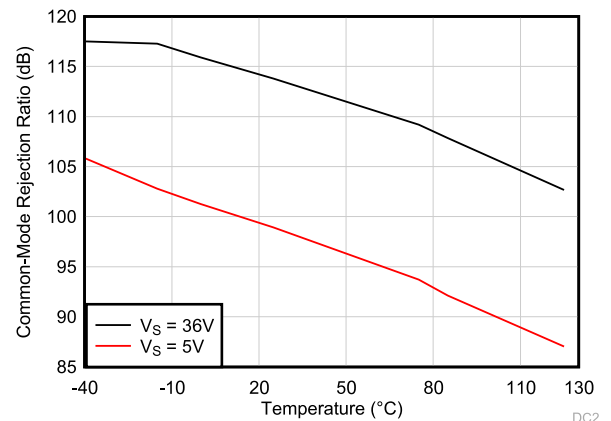


Figure 5-14. Common-Mode Rejection Ratio vs Temperature (dB)

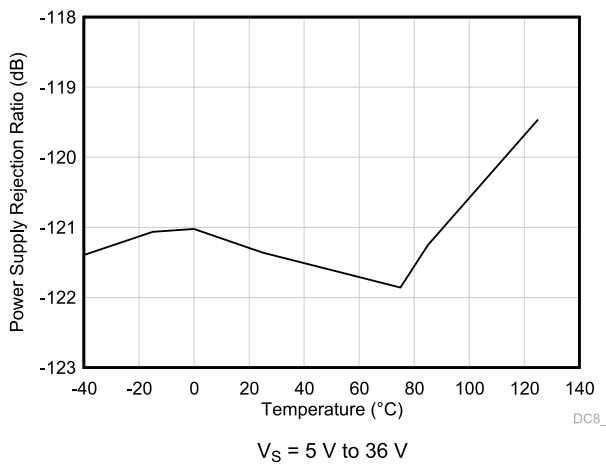


Figure 5-15. Power Supply Rejection Ratio vs Temperature (dB)

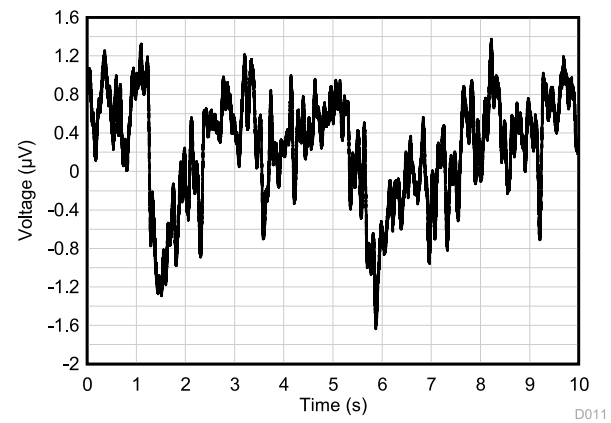


Figure 5-16. 0.1-Hz to 10-Hz Noise

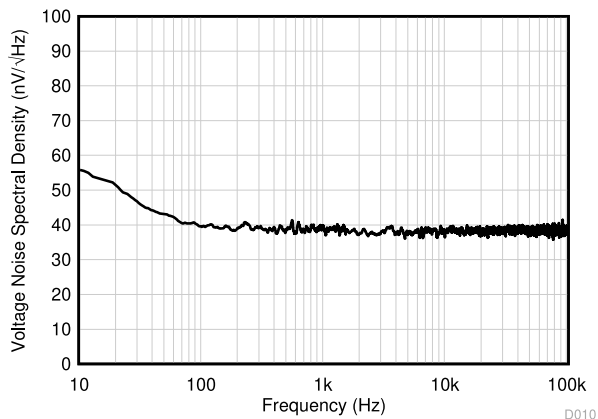


Figure 5-17. Input Voltage Noise Spectral Density vs Frequency

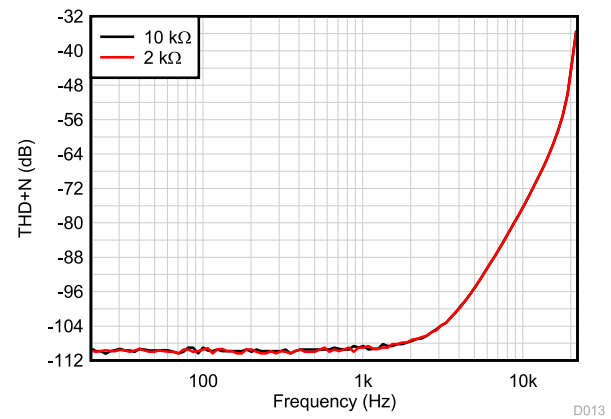


Figure 5-18. THD+N Ratio vs Frequency, $G = 1$
 $G = 1$, $f = 1\text{ kHz}$, $BW = 80\text{ kHz}$,
 $V_{OUT} = 10\text{ V}_{PP}$, R_L connected to V_-

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

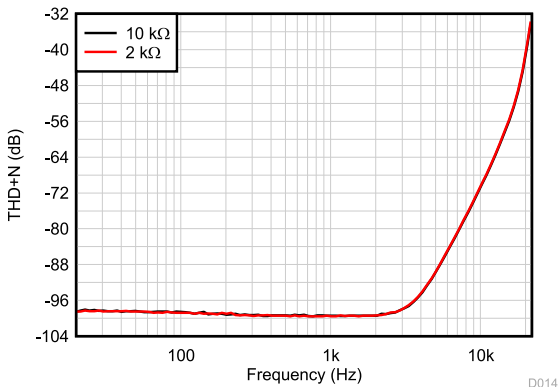


Figure 5-19. THD+N Ratio vs Frequency, $G = -1$

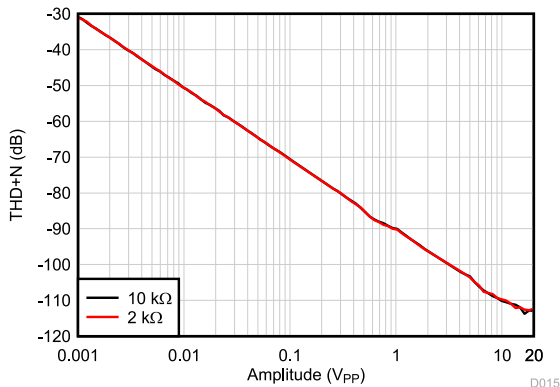


Figure 5-20. THD+N vs Output Amplitude, $G = 1$

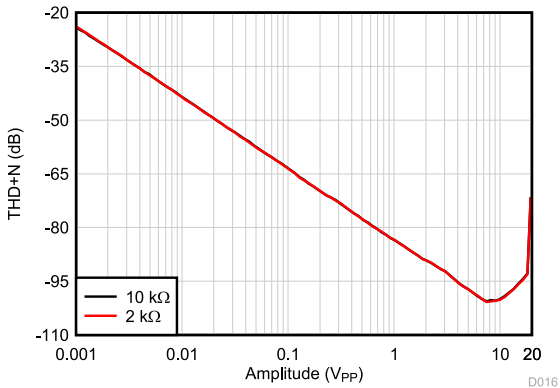


Figure 5-21. THD+N vs Output Amplitude, $G = -1$

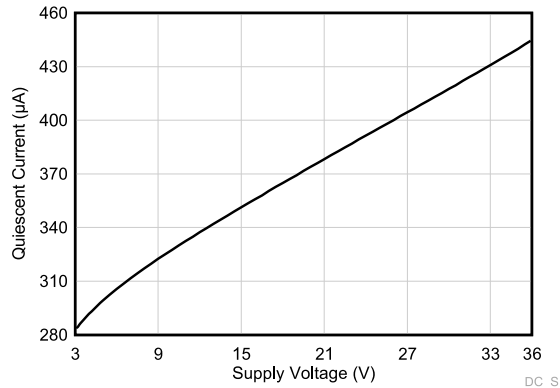


Figure 5-22. Quiescent Current vs Supply Voltage

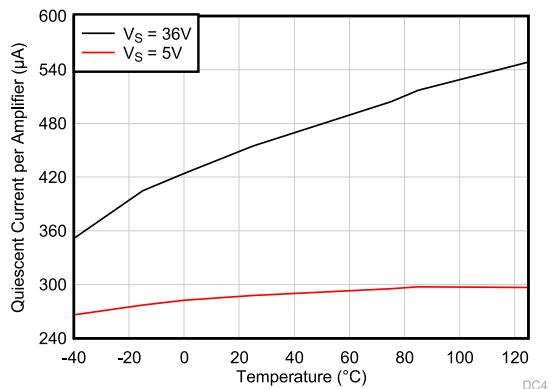


Figure 5-23. Quiescent Current vs Temperature

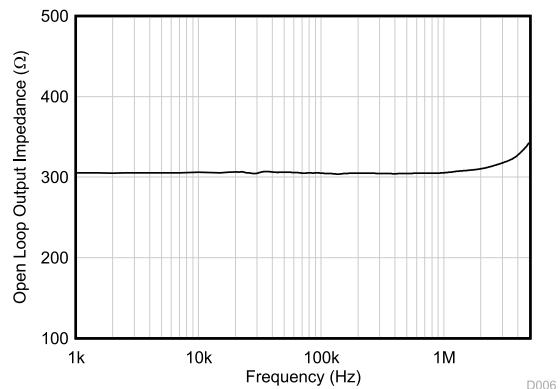


Figure 5-24. Open-Loop Output Impedance vs Frequency

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

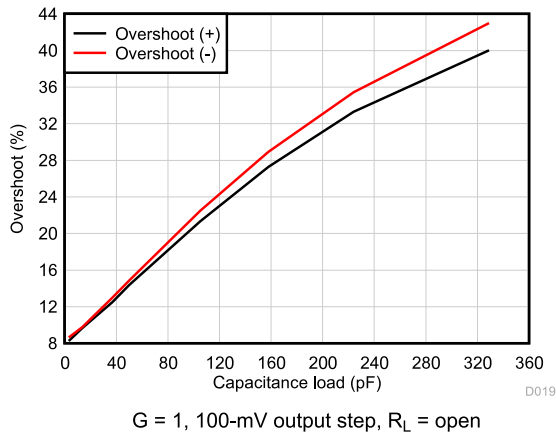


Figure 5-25. Small-Signal Overshoot vs Capacitive Load

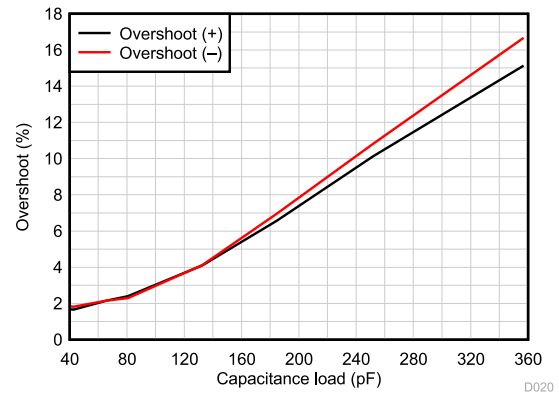


Figure 5-26. Small-Signal Overshoot vs Capacitive Load

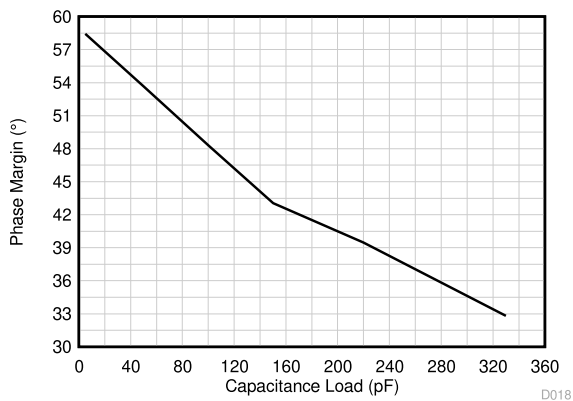


Figure 5-27. Phase Margin vs Capacitive Load

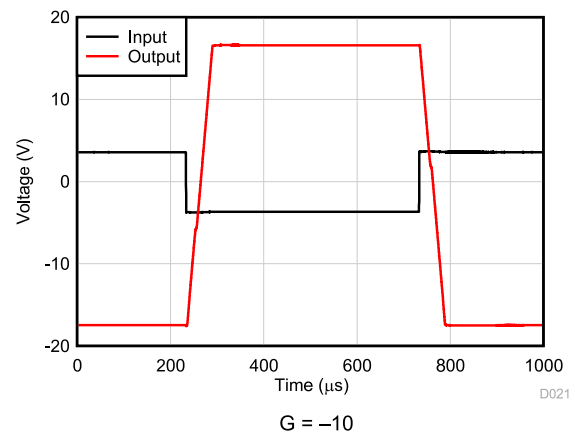


Figure 5-28. Overload Recovery

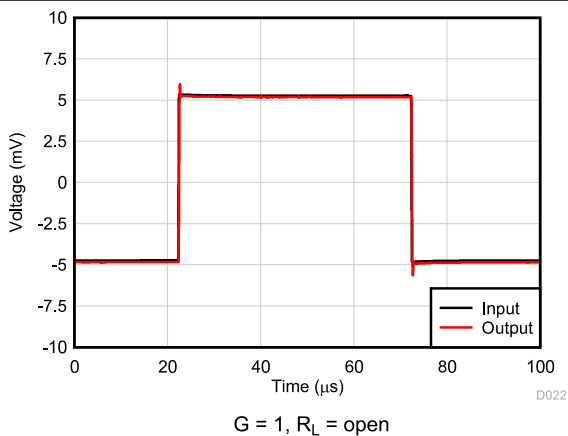


Figure 5-29. Small-Signal Step Response, $G = 1$

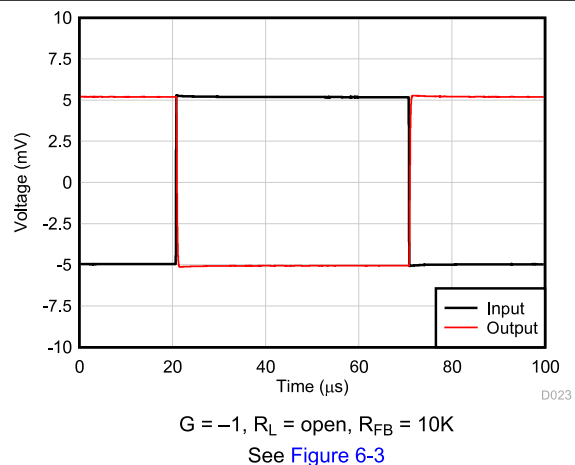


Figure 5-30. Small-Signal Step Response, $G = -1$

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

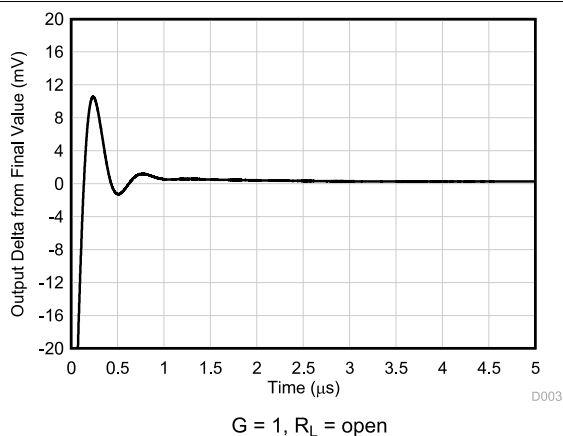


Figure 5-31. Large-Signal Step Response (Rising)

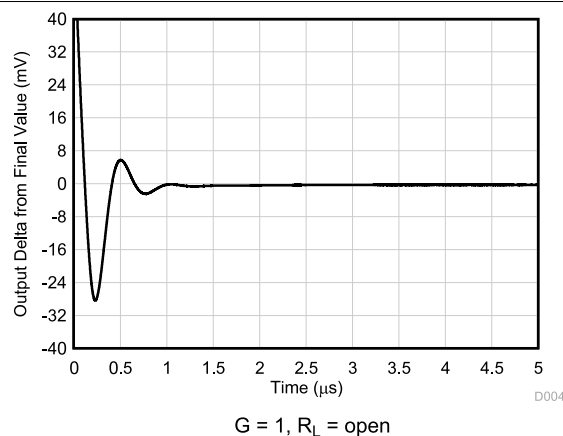


Figure 5-32. Large-Signal Step Response (Falling)

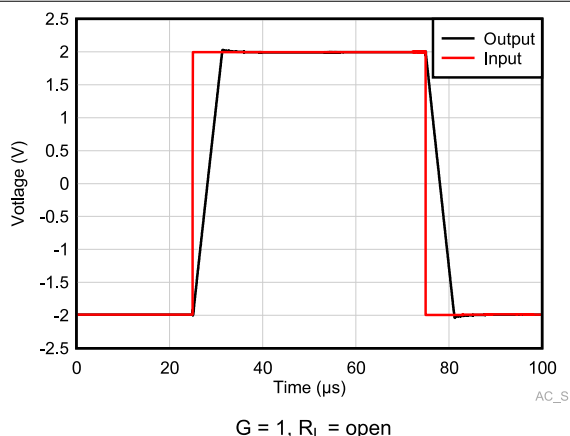


Figure 5-33. Large-Signal Step Response

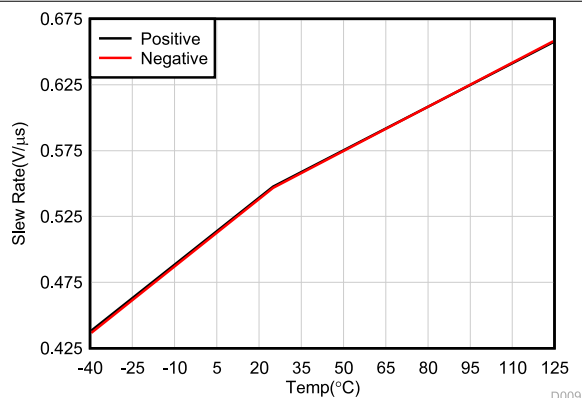


Figure 5-34. Slew Rate vs Temperature

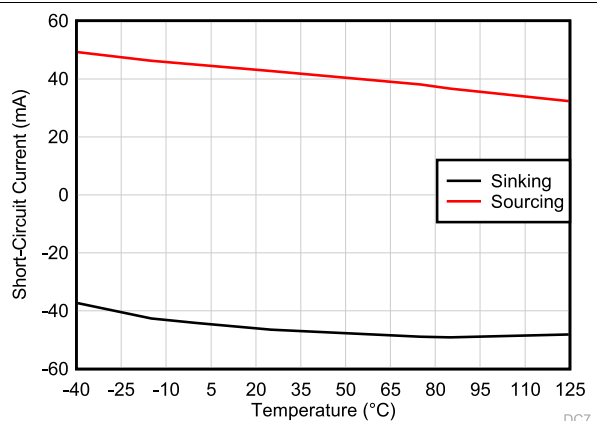


Figure 5-35. Short-Circuit Current vs Temperature

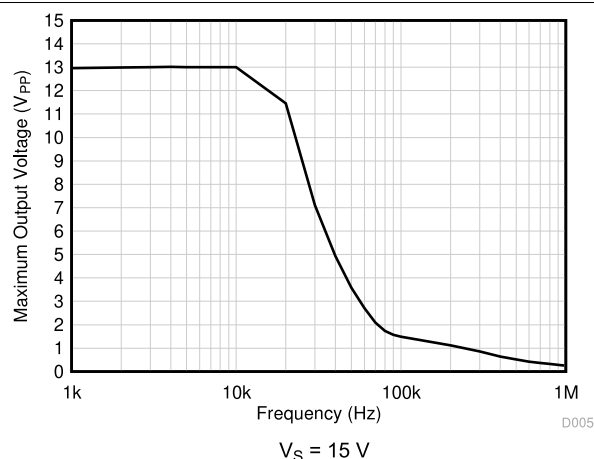


Figure 5-36. Maximum Output Voltage vs Frequency

5.11 Typical Characteristics: LM358B and LM2904B (continued)

This typical characteristics section is applicable for LM358B and LM2904B. Typical characteristics data in this section was taken with $T_A = 25^\circ\text{C}$, $V_S = 36\text{ V}$ ($\pm 18\text{ V}$), $V_{CM} = V_S / 2$, $R_{LOAD} = 10\text{ k}\Omega$ connected to $V_S / 2$ (unless otherwise noted).

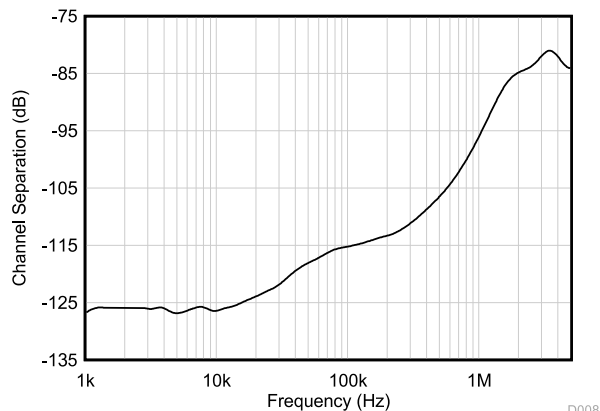


Figure 5-37. Channel Separation vs Frequency

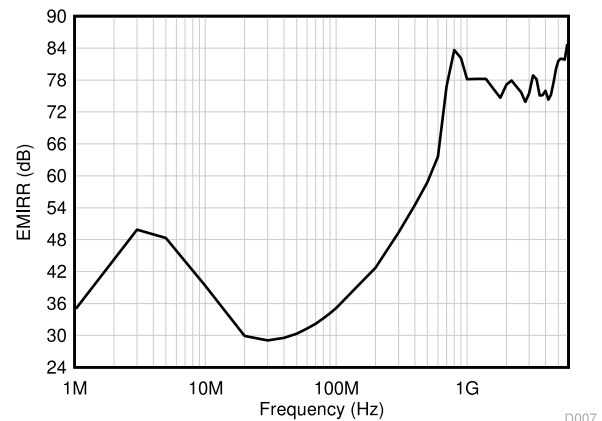


Figure 5-38. EMIRR (Electromagnetic Interference Rejection Ratio) vs Frequency

5.12 Typical Characteristics: LM158, LM158A, LM258, LM258A, LM358, LM358A, LM2904, and LM2904V

Typical characteristics section is applicable for LM158, LM158A, LM258, LM258A, LM358, LM358A, LM2904, and LM2904V.

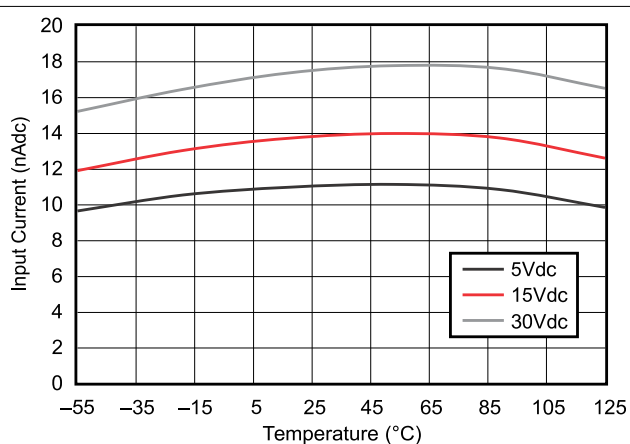


Figure 5-39. Input Current vs Temperature

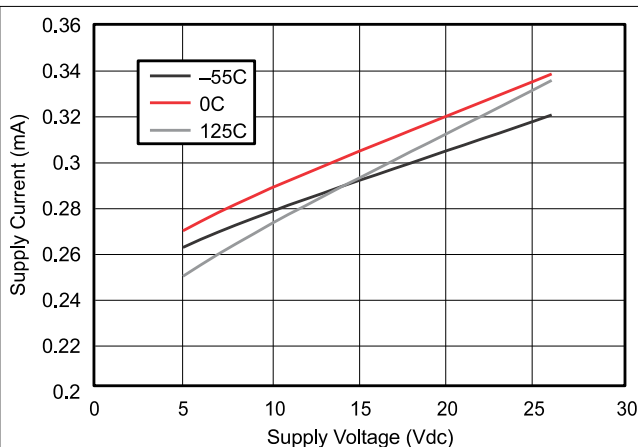


Figure 5-40. Supply Current vs Supply Voltage

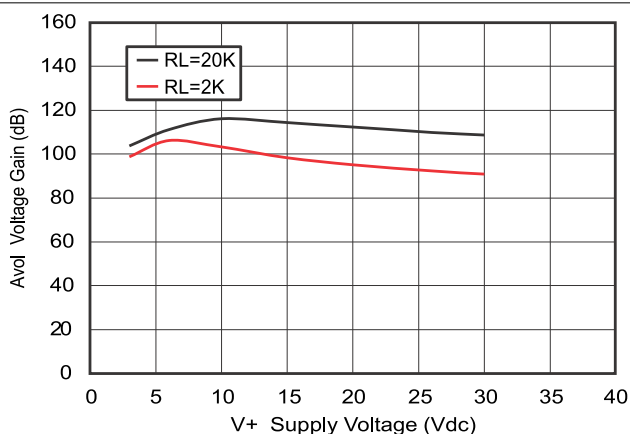


Figure 5-41. Voltage Gain vs Supply Voltage

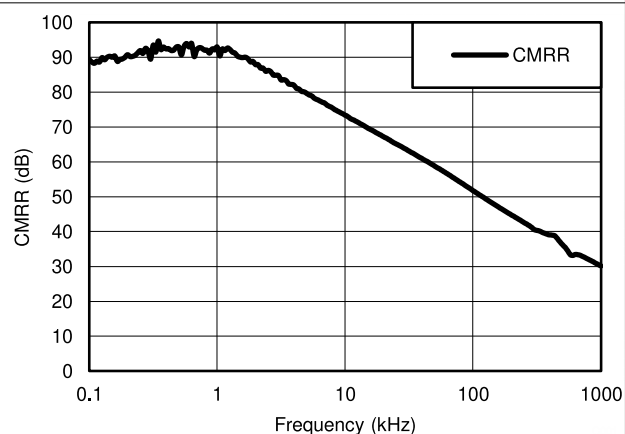


Figure 5-42. Common-Mode Rejection Ratio vs Frequency

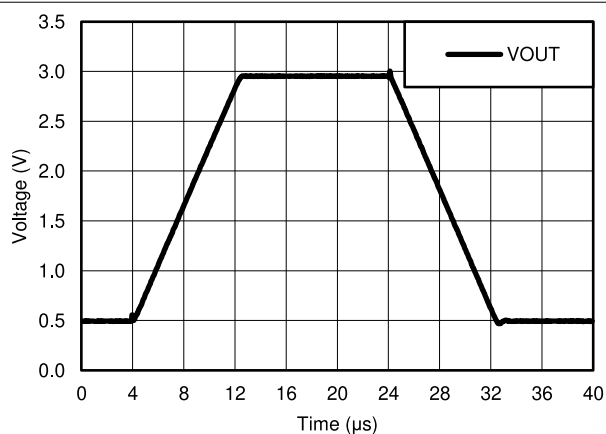


Figure 5-43. Voltage Follower Large Signal Response (50 pF)

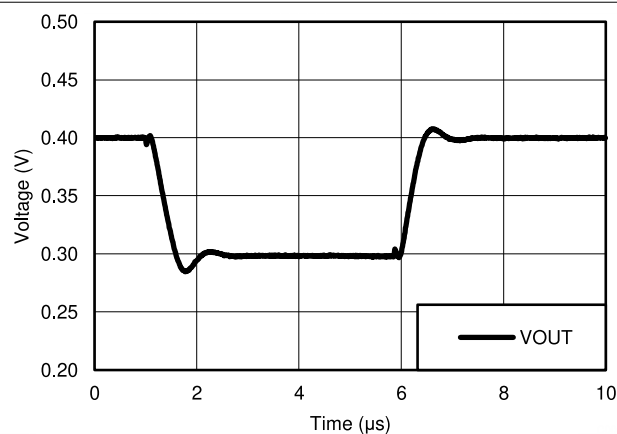


Figure 5-44. Voltage Follower Small Signal Response (50 pF)

5.12 Typical Characteristics: LM158, LM158A, LM258, LM258A, LM358, LM358A, LM2904, and LM2904V (continued)

Typical characteristics section is applicable for LM158, LM158A, LM258, LM258A, LM358, LM358A, LM2904, and LM2904V.

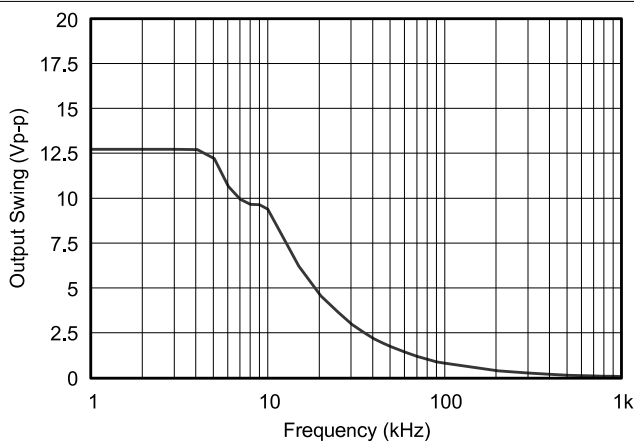


Figure 5-45. Maximum Output Swing vs Frequency ($V_{CC} = 15\text{ V}$)

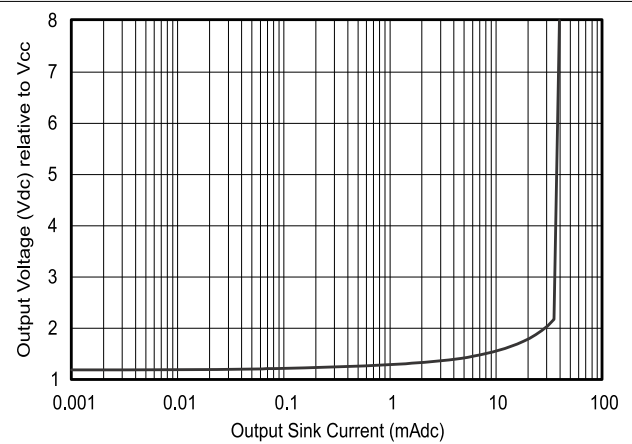


Figure 5-46. Output Sourcing Characteristics

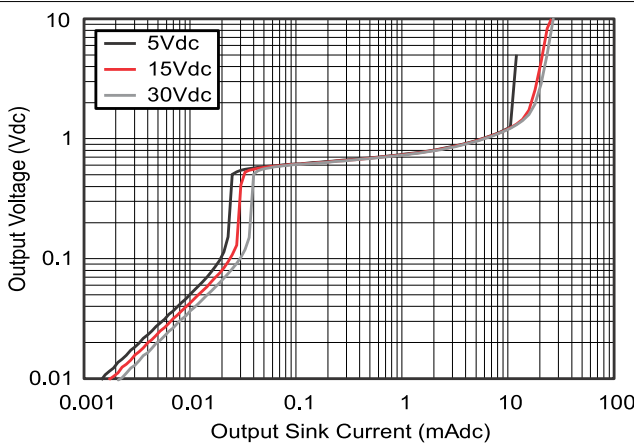


Figure 5-47. Output Sinking Characteristics

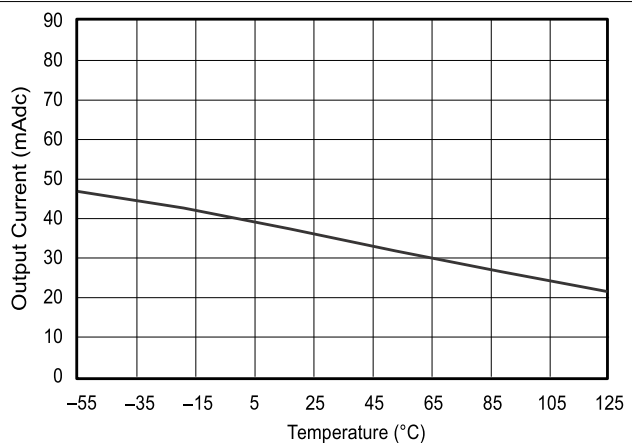


Figure 5-48. Source Current Limiting